# Case Study II: Styrian Health Data

Week 3 Update: Modelling Improvements

Rohith B Chowdappa
Nixon M Paulson
Maria Cherian

# Focus of This Week

- Continued from last week's EDA and initial Random Forest modelling
- Refined the model based on feedback from the previous presentation
- Focused on multi-output blood pressure prediction
- Added temporal and weather-based feature engineering
- Extended the work with classification and residual diagnostics

# Feedback Addressed

- Moved away from relying only on the default train-test split
- Focused on the multi-output model instead of separate models only
- Clarified that feature importance is predictive, not causal
- Explored temporal and environmental effects
- Added classification to compare exact BP prediction with risk-category prediction

# Updated Modelling Pipeline

- Train-test split was stratified by age groups
- Random Forest depth was tuned to reduce overfitting
- Cross-validation was used to check model stability
- New temporal variables:
  - hour, weekday, month, season
- External weather variables:
  - temperature and humidity, matched by measurement hour

# Model Robustness Checks

**5-fold CV, baseline RF**

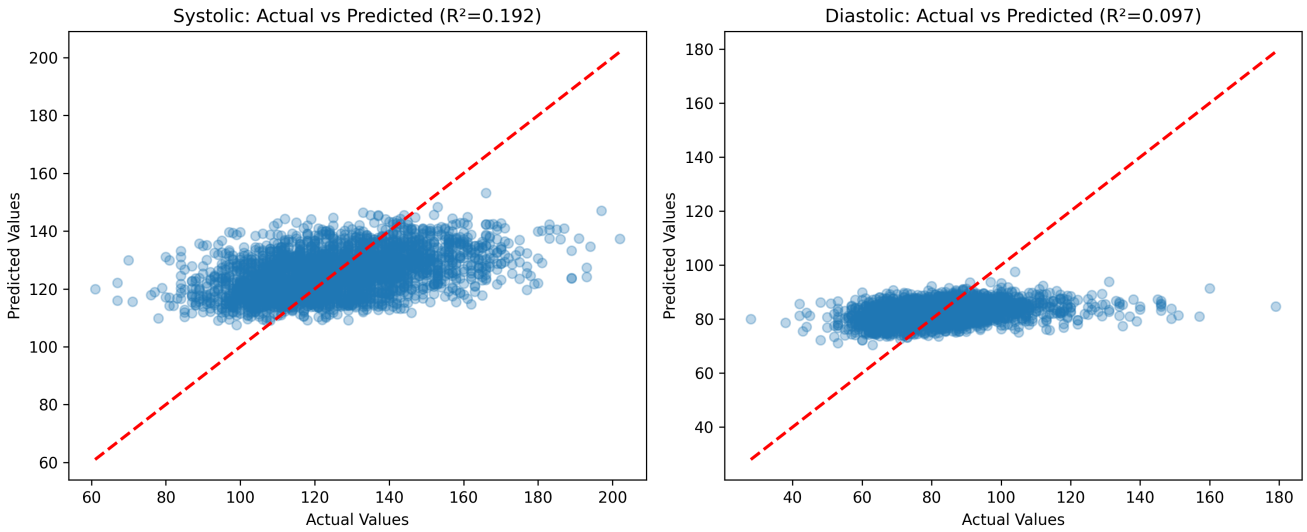
| Target | Mean $R^2$ | Std. dev. |
|---|---|---|
| Systolic | 0.140 | 0.013 |
| Diastolic | 0.035 | 0.014 |

**Tree depth experiment**

| Max depth | Mean $R^2$ |
|---|---|
| 3 | 0.138 |
| 5 | 0.157 |
| 10 | 0.163 |
| 20 | 0.140 |
| None | 0.140 |

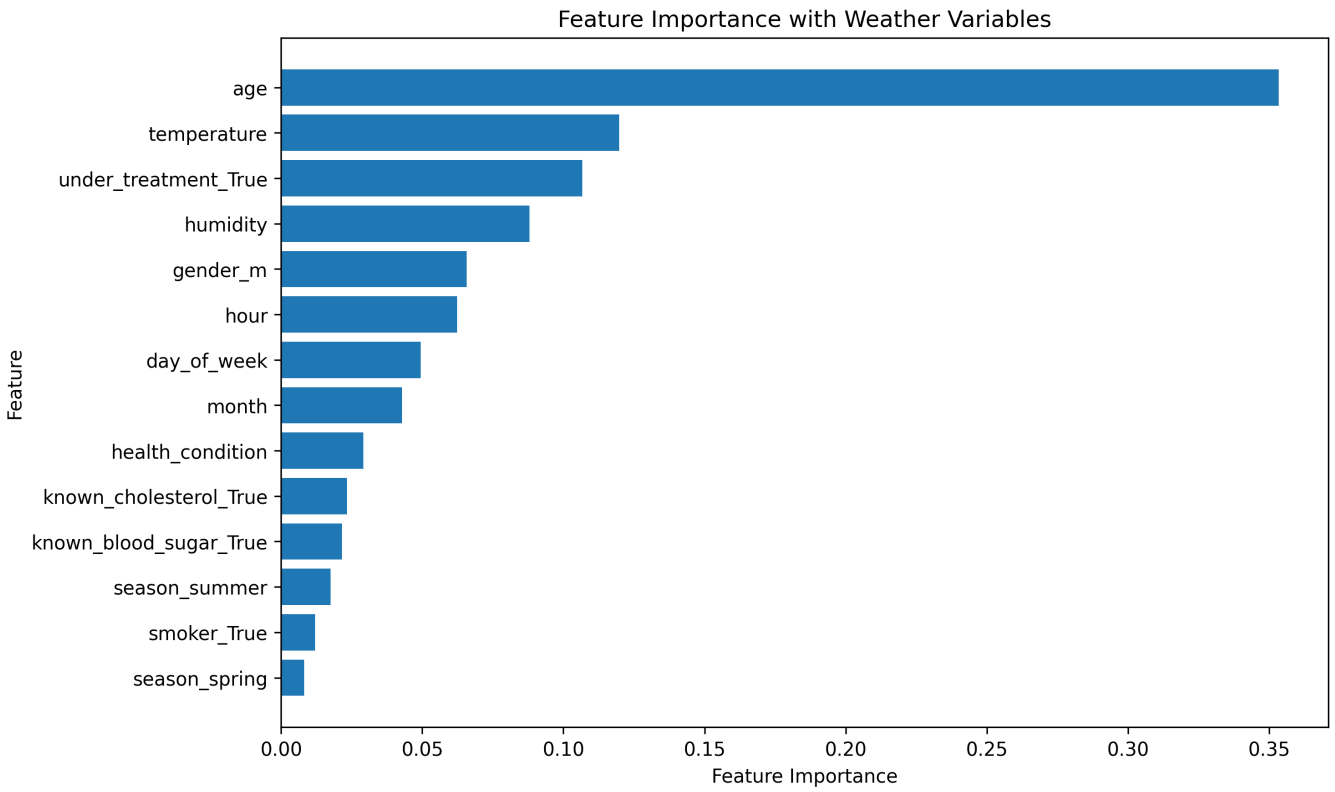
- Moderate tree depth performed best
- Very deep trees showed signs of overfitting

# Regression Performance Improvement



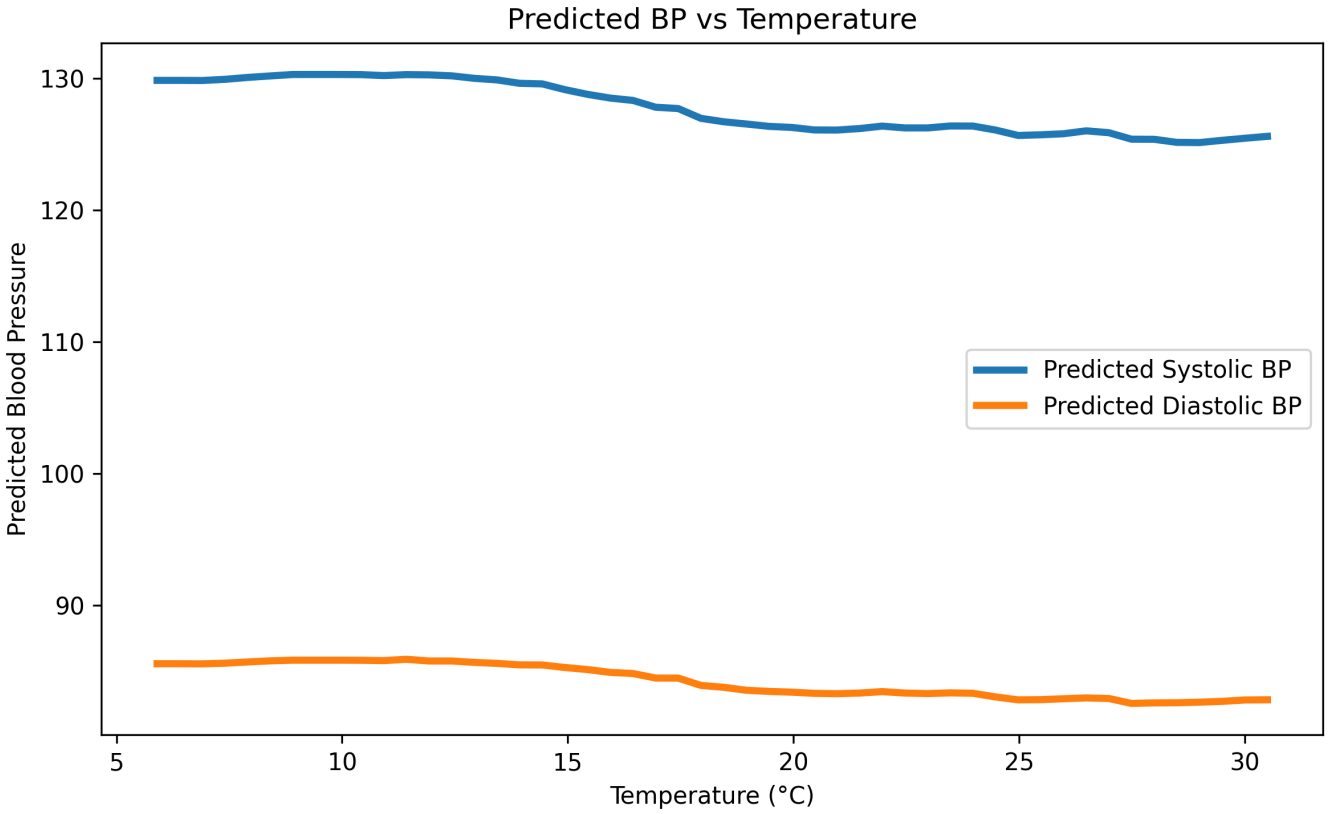

Adding temporal and weather features improved prediction for both systolic and diastolic BP. The improvement is meaningful, although individual-level BP prediction remains difficult.

# Feature Importance with Weather Variables



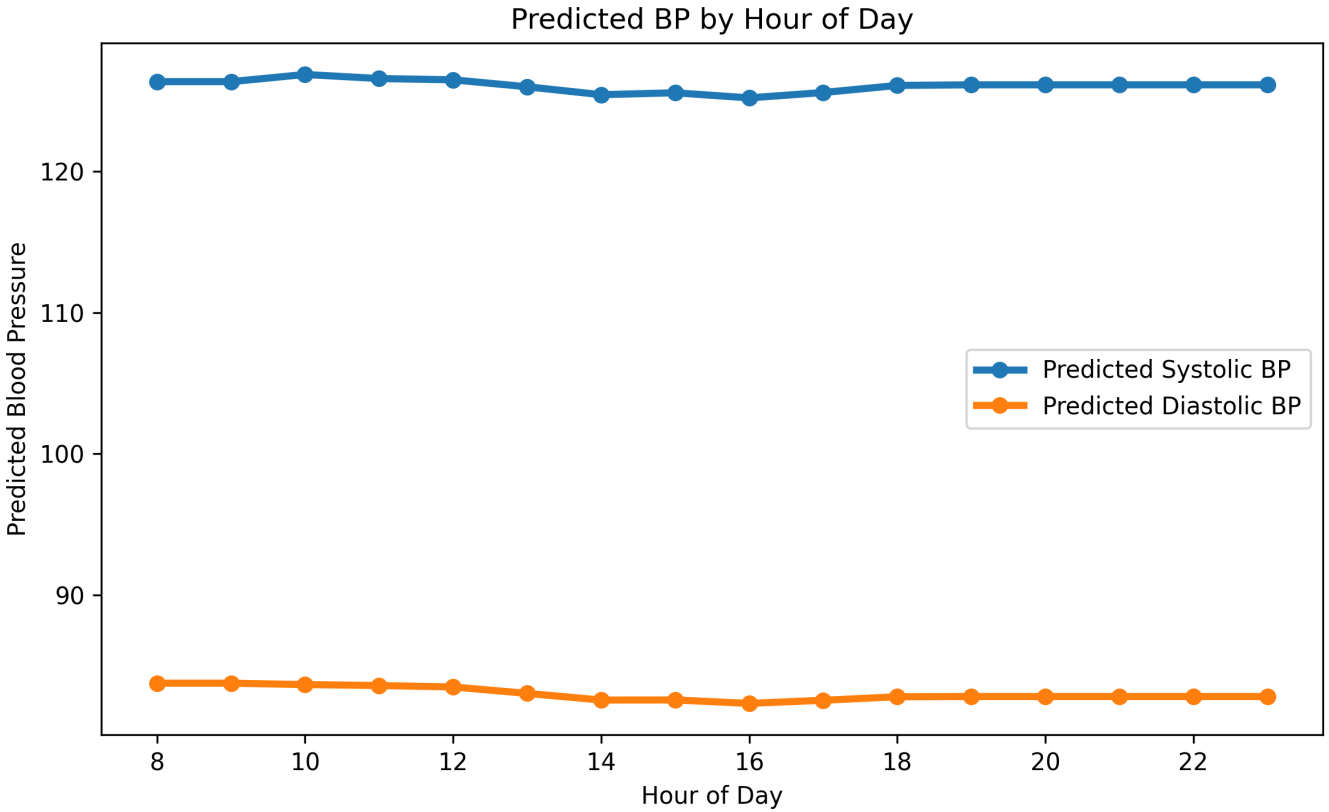

Age remained the strongest predictor, but temporal and environmental variables also contributed predictive information.

# Predicted BP vs Temperature



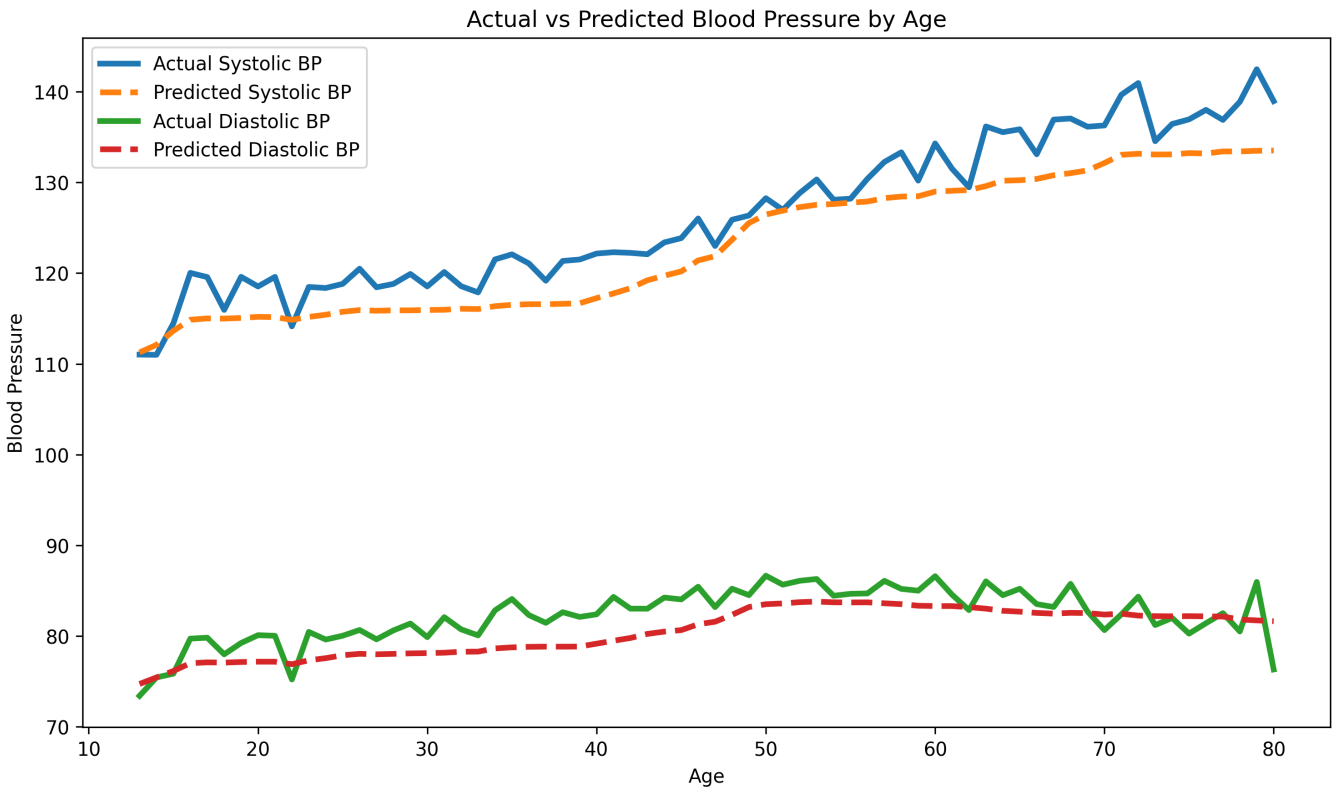

The weather-enhanced model learned a weak inverse association between temperature and predicted BP. This should be interpreted as predictive association, not causality.

# Predicted BP by Hour of Day



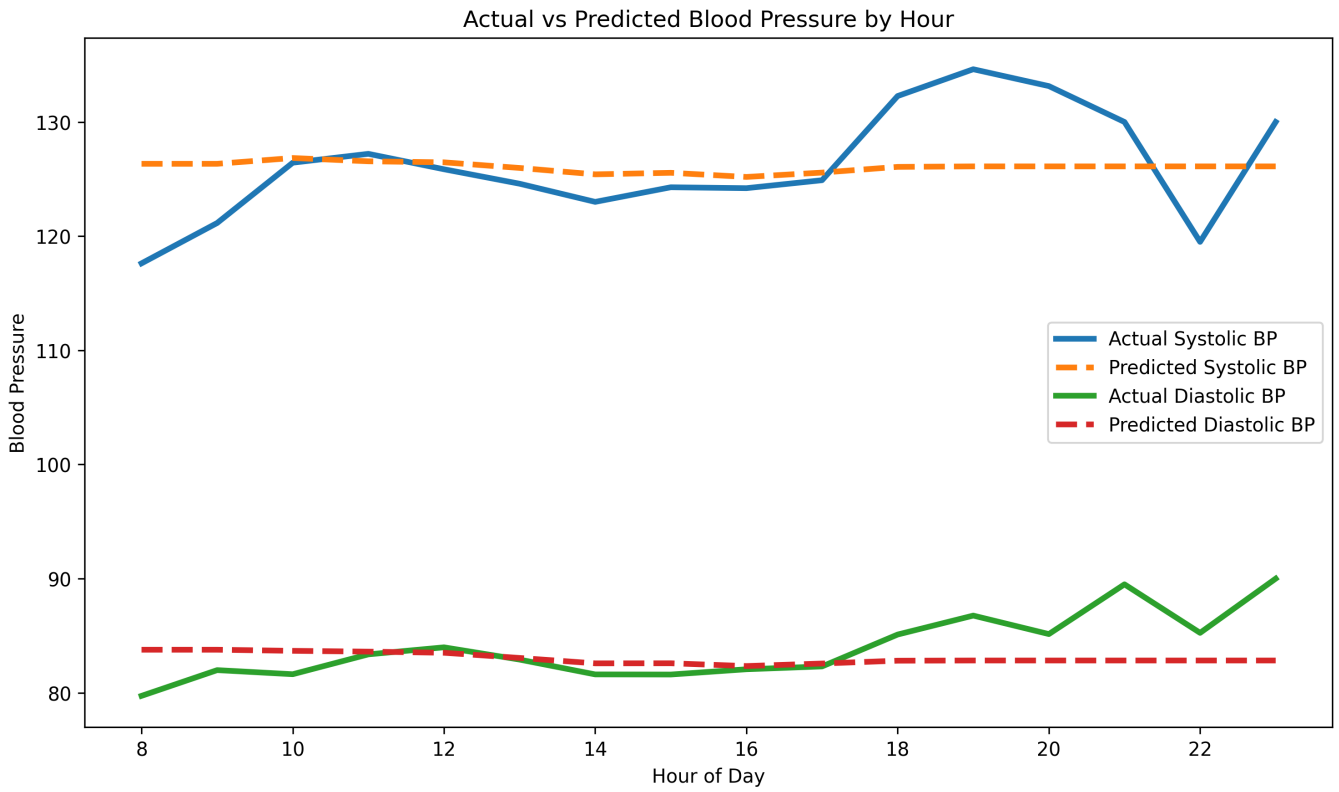

The model learned modest within-day variation. Hour alone does not dominate prediction, but it adds useful temporal context.

# Actual vs Predicted BP by Age



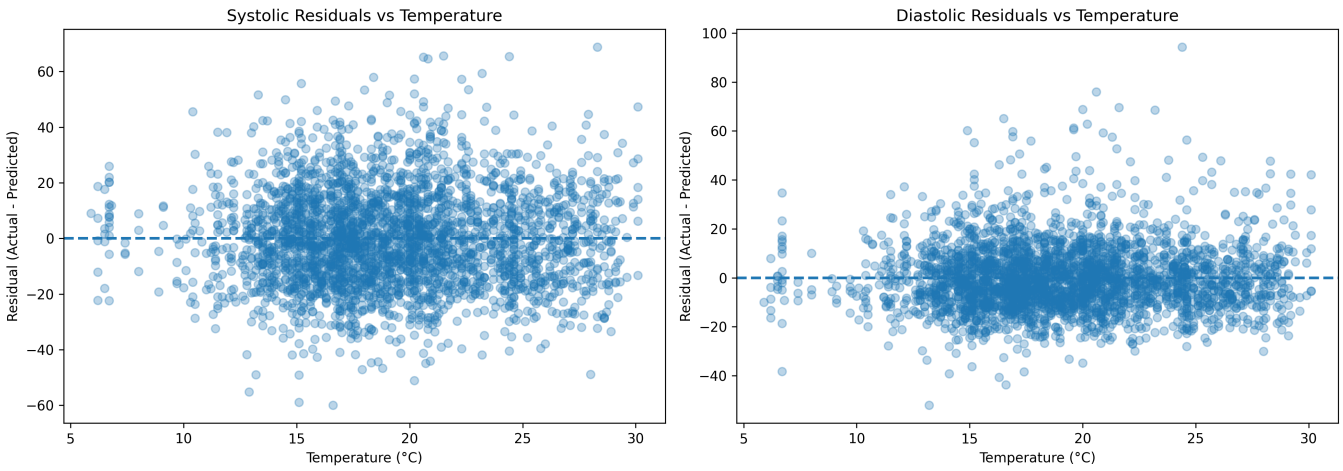

The model captures broader age-related trends, especially for systolic BP, while smoothing noisy individual variation.

# Actual vs Predicted BP by Hour




The model captures the general hourly level, but sharp short-term fluctuations remain difficult to predict.

# Residual Diagnostics




Residuals remain roughly centered across temperature values, suggesting no strong systematic temperature-related prediction bias.

# Classification Extension

- Regression predicts exact BP values
- Classification predicts elevated BP categories
- Systolic high BP:

$$BP_{sys} \geq 140$$

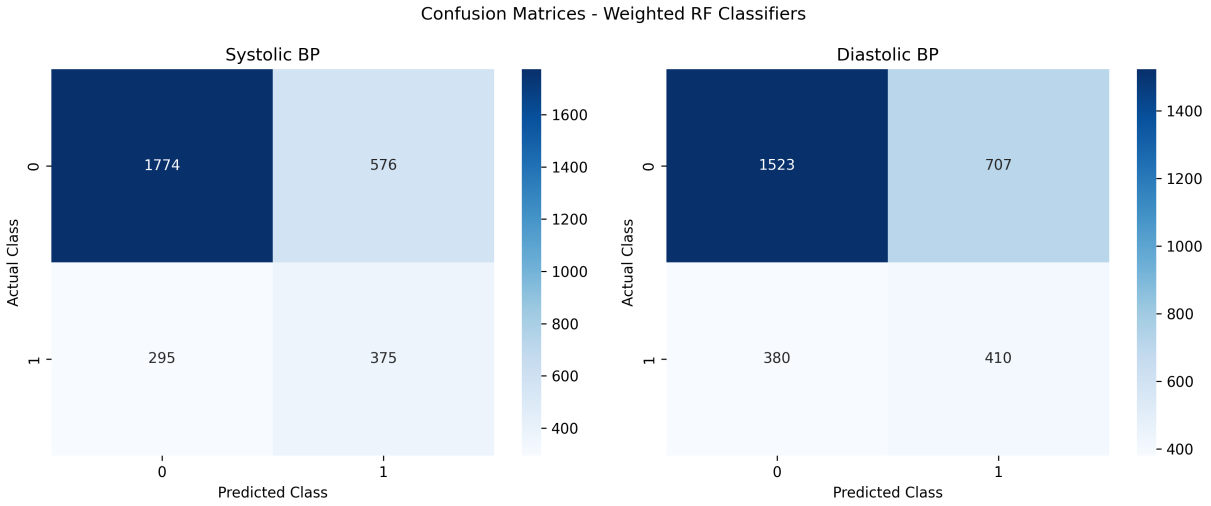
- Diastolic high BP:

$$BP_{dia} \geq 90$$

- Weighted Random Forest was used because high BP cases were less frequent

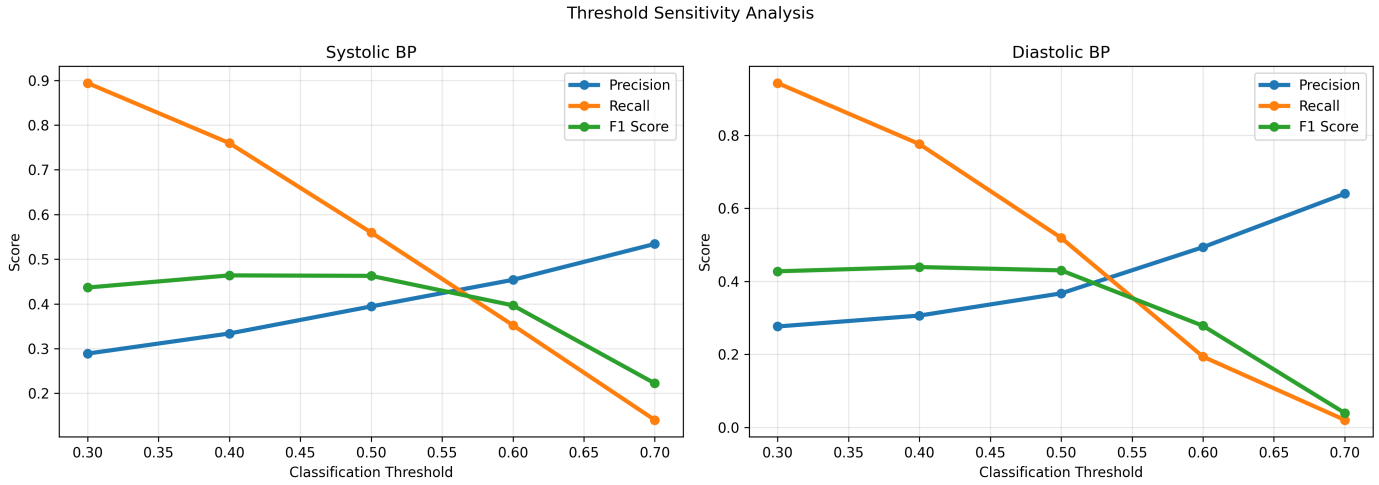
# Classification Results

| Target | Accuracy | Precision | Recall | F1 |
|---|---|---|---|---|
| Systolic BP | 0.712 | 0.394 | 0.560 | 0.463 |
| Diastolic BP | 0.640 | 0.367 | 0.519 | 0.430 |




Systolic classification performed slightly better, but both targets show a clear precision–recall tradeoff.

# Threshold Sensitivity Analysis




Lower thresholds increase recall and detect more elevated BP cases, while higher thresholds reduce false positives.

# Current Takeaways and Open Questions

- Temporal and weather features improved regression performance
- The model captures population-level trends better than individual values
- Classification gives a useful risk-category perspective
- Threshold choice strongly affects false positives and false negatives
- Open question: should final focus be regression performance, classification, or model interpretation?